

1 Summary

Model (Run-name): model1 (test)

Iteration: 32

TAC: 154888.34876031912

Took: 3.7562577724456787s

Total Time: 66.5010313987732s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

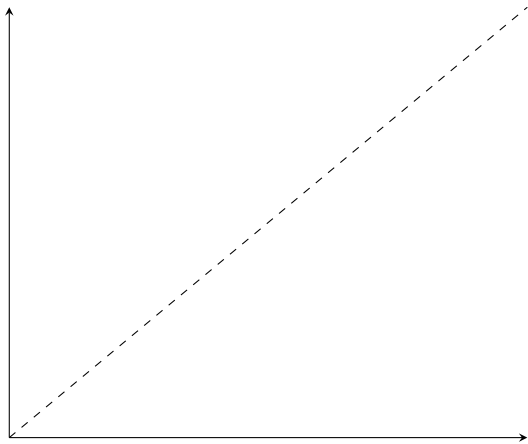
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

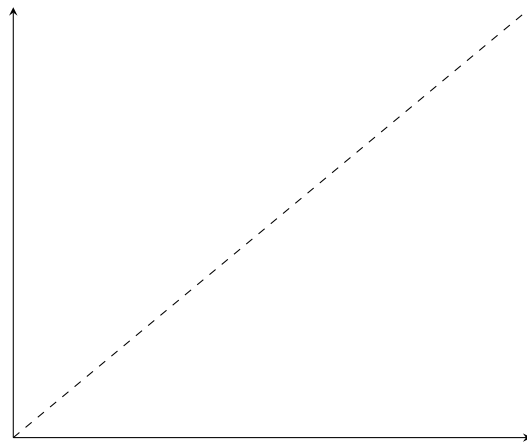
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.051989435561420536	$b_{1,1,1}^{H,out}$	908.4103244246526
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.017932456215420663	$b_{1,2,2}^{H,out}$	3415.149664989693
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.028466884276826736	$b_{2,1,2}^{H,out}$	7868.75389751948
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014681836407067606	$b_{1,1,1}^{C,in}$	6869.247836650746
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009560548177268806	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022499980559771616	$b_{2,1,2}^{C,in}$	5833.652498156983
$A_{1,1,1}^\beta$	71.3743618177527	$q_{1,1,1}$	687.4456509159542	$b_{1,1,1}^{C,out}$	7556.6934875667
$A_{1,2,2}^\beta$	69.93657924014246	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	137.63821724432228	$q_{2,1,2}$	2418.8315236260664	$b_{2,1,2}^{C,out}$	8252.48402178305
$A_{cu,1}^\beta$	4.75409580038526	$q_{cu,1}$	162.5543490840272		
$A_{cu,2}^\beta$	37.87287854392874	$q_{cu,2}$	1981.1684763739358		
$A_{hu,1}^\beta$	13.321736298873567	$q_{hu,1}$	493.72282545798646		
$A_{1,1,1}$	71.3743618177527	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	69.93657924014246	$t_{1,2}^{(H)}$	581.2554349084035		
$A_{2,1,2}$	137.63821724432228	$t_{1,3}^{(H)}$	386.2554349084027		
$A_{cu,1}$	4.75409580038526	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.87287854392874	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.321736298873567	$t_{2,3}^{(H)}$	469.05842381869684		
$dt_{1,1,1}$	32.914855030532564	$t_{1,1}^{(C)}$	617.0851449694676		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	571.2554349084041		
$dt_{1,2,2}$	81.25543490840272	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	36.25543490840271	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	18.744565091596428	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	59.058423818696596	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	66.25543490840272	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	149.05842381869684	$t_{1,2,2}^H$	370.00557530345407		
$dt_{hu,1}$	62.91485503053243	$t_{2,1,2}^H$	451.27837189025587		
$f_{1,1,1}^H$	2.4551630389855488	$t_{1,1,1}^C$	626.4589713847378		
$f_{1,2,2}^H$	9.230134229701937	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	17.436585459568718	$t_{2,1,2}^C$	579.5111508959654		
$f_{1,1,1}^C$	12.025397342001094	$b_{1,1,1}^{H,in}$	1595.8559753406066		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5365.1496649896935		
$f_{2,1,2}^C$	14.228420727212153	$b_{2,1,2}^{H,in}$	10287.585421145546		

3 *RecLMTD* Tangent Points

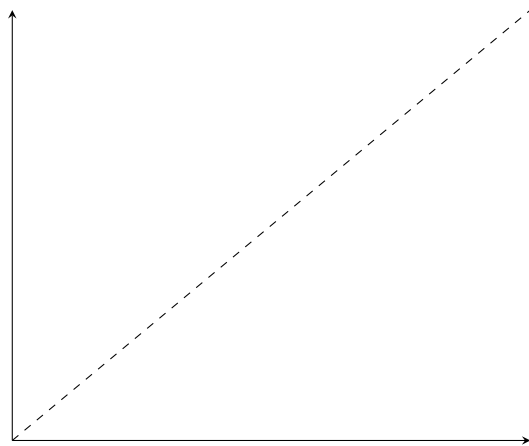
- $(1, 1, 1)$
 - Value: 0.051989
 - Error: 0.000000375500879
 - Relative error: 0.000007222585950
- $(1, 2, 2)$
 - Value: 0.017932
 - Error: 0.000001064041977
 - Relative error: 0.000059332577289
- $(2, 1, 2)$
 - Value: 0.028467
 - Error: 0.000000332104376
 - Relative error: 0.000011666204777
- $cu, 1$
 - Value: 0.014682
 - Error: 0.000000174146827
 - Relative error: 0.000011861238371
- $cu, 2$
 - Value: 0.009561
 - Error: 0.000000015475526
 - Relative error: 0.000001618683483
- $hu, 1$
 - Value: 0.022500
 - Error: 0.000000040250912
 - Relative error: 0.000001788927755



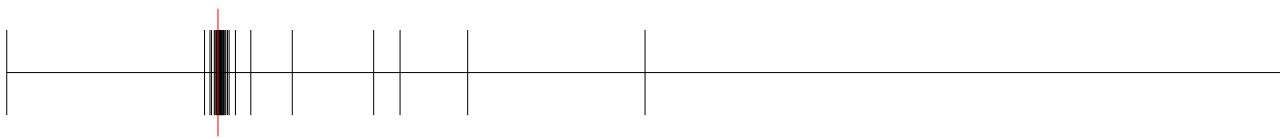
(a) *RecLMTD* - (1, 1, 1)



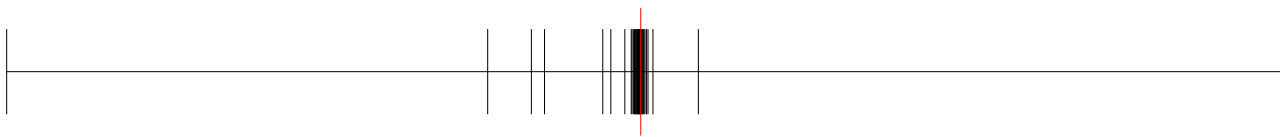
(b) *RecLMTD* - (1, 2, 2)



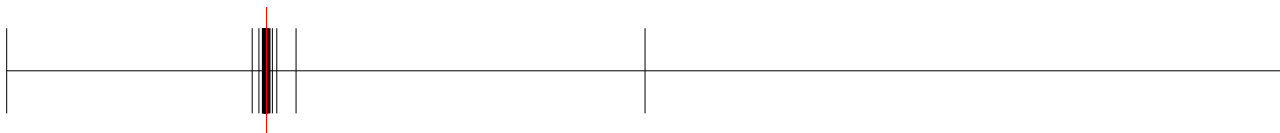
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - cu, 2



(a) *RecLMTD* - hu, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1595.8559753406066
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5365.1496649896935
 - Absolute error: 0.08397904135290446
 - Relative error: 1.565293815001263e-05
- (2, 1, 2)
 - Value: 10287.585421145546
 - Absolute error: 1.8189894035458565e-12
 - Relative error: 1.768140267206946e-16

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 908.4103244246526
 - Absolute error: 4.547473508864641e-13
 - Relative error: -5.005968543724787e-16
- (1, 2, 2)
 - Value: 3415.149664989693
 - Absolute error: 0.05146079927590108
 - Relative error: -1.5068160667700873e-05
- (2, 1, 2)
 - Value: 7868.75389751948
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 6869.247836650746
 - Absolute error: 0.32575190045645286
 - Relative error: -4.7419522661399154e-05
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 5833.652498156983
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 7556.6934875667
 - Absolute error: 23.275438203935664
 - Relative error: 0.0030896251942243514
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 8252.48402178305
 - Absolute error: 6.9555507243258035
 - Relative error: 0.0008435542668658947



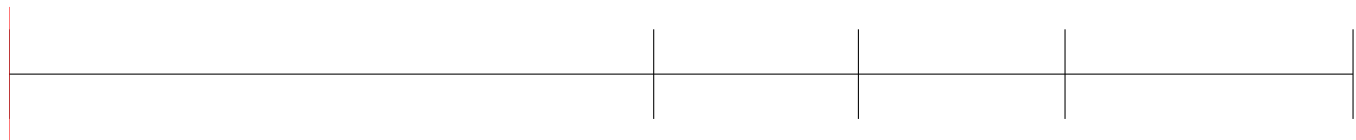
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



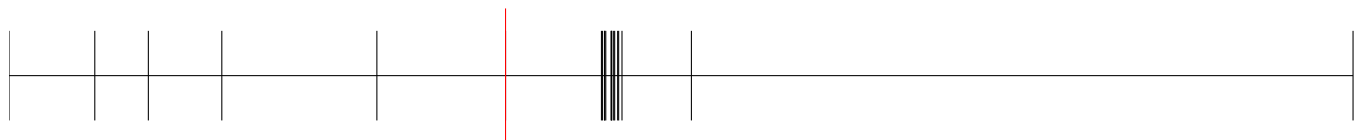
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



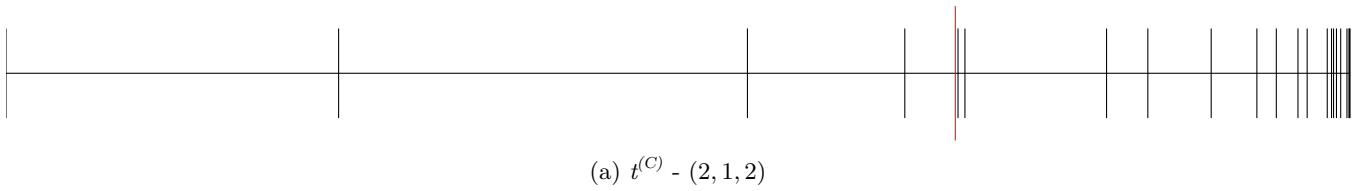
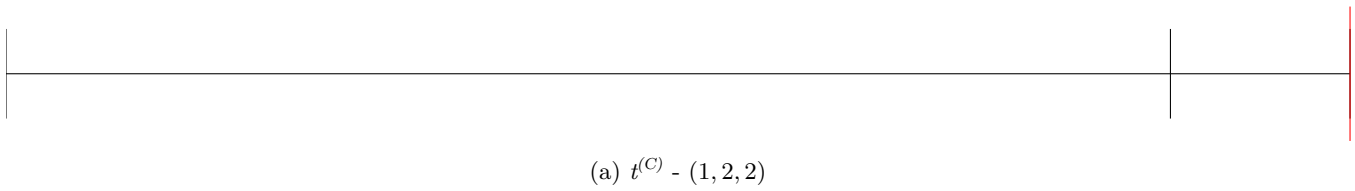
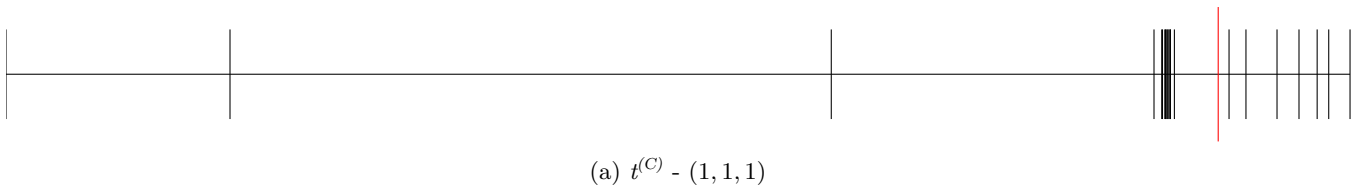
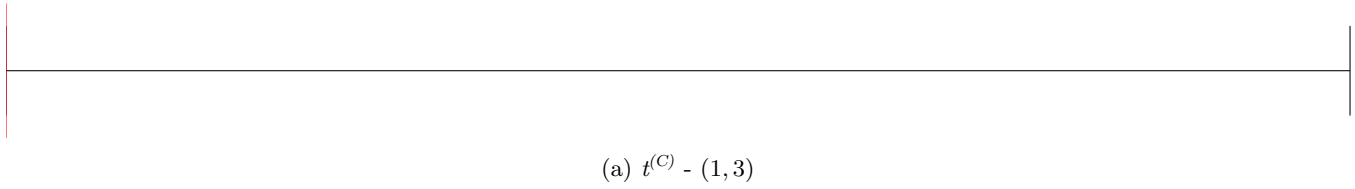
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

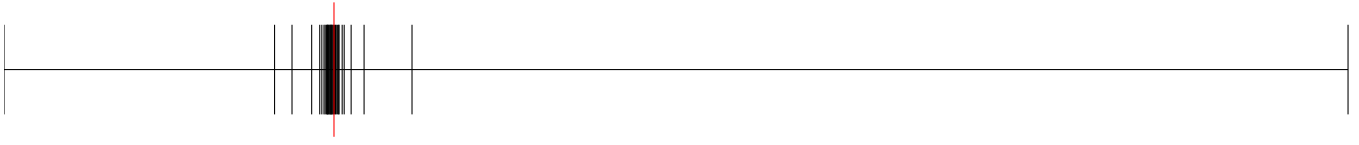
- $(1, 1, 1)$
 - Value: 71.374362
 - Error: 0.105460922794890
 - Relative error: 0.001475394296621
- $(1, 2, 2)$
 - Value: 69.936579
 - Error: 0.000000000001876
 - Relative error: -0.000000000000027
- $(2, 1, 2)$
 - Value: 137.638217
 - Error: 0.074976892085175
 - Relative error: 0.000544442328532

5.2 A_{CU}

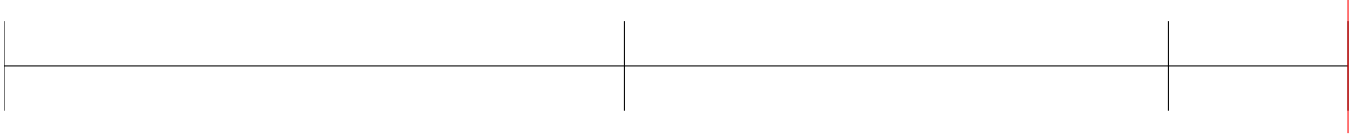
- $cu, 1$
 - Value: 4.754096
 - Error: 0.019096920632834
 - Relative error: 0.004000869386384
- $cu, 2$
 - Value: 37.872879
 - Error: 0.009234787389758
 - Relative error: 0.000243776985433

5.3 A_{HU}

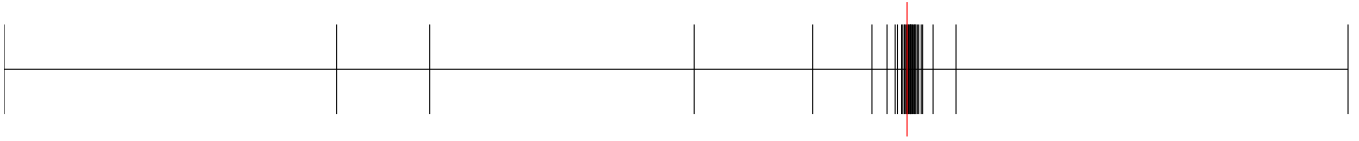
- $hu, 1$
 - Value: 13.321736
 - Error: 0.008768470790679
 - Relative error: 0.000657774851154



(a) $A^\beta - (1, 1, 1)$



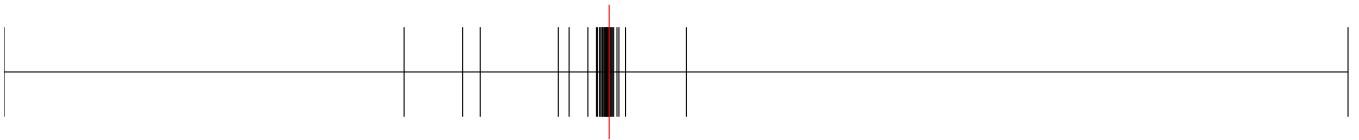
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

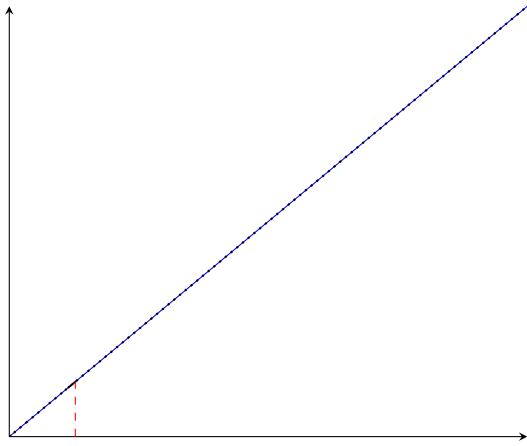


(a) $A^\beta - hu, 1$

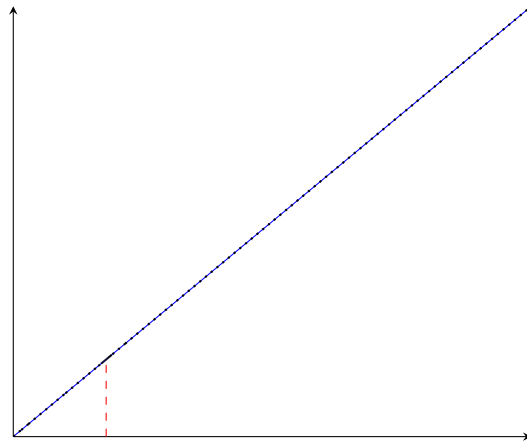
6 A^β Breakpoints

- (1, 1, 1)
 - q : 71.374362
 - $\hat{q}^\beta$: 71.374362
 - q^β : 71.374362
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.49758876855047, 69.89902342903967, 70.11420238741596, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, 70.58322282660124, 70.60541450524272, 70.65581083979713, 70.83181816751056, 70.95870595008942, 71.00935653192329, 71.03465240716494, 71.18527282143235, 71.22933662920877, 71.2616536557244, **71.33750386361461**, **71.62767096549581**, 71.6637831223933, 71.88123217857378, 72.01156079815767, 72.36161285231599, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 69.936579
 - $\hat{q}^\beta$: 69.936579
 - q^β : 69.936579
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 66.74637189255778, 67.27673099173637, 67.60071888621576, 67.91344539650363, 68.10969007738701, 68.20353230192663, 68.48329505760876, 68.66676327863097, 68.85281320938876, 69.0314457108091, 69.13316279873226, 69.22655548799992, 69.34502840170047, 69.41355730879985, **69.60484346150977**, **70.08419437796795**, 70.37078617087542, 70.56626459266522, 70.94081287188642, 71.26369181826138, 72.25348351595485, 72.87506706799363, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 137.638217
 - $\hat{q}^\beta$: 137.638217
 - q^β : 137.638217
 - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 127.07690270456857, 128.52021081477858, 131.69760457629607, 132.12228195956155, 132.5928424015674, 135.25996987589937, 135.81963163113545, **137.02923097052596**, **138.86359281280824**, 139.47623233772285, 140.31685261310218, 140.7008934199032, 140.73764915547463, 141.79783607898793, 142.4338700260237, 143.30274961189664, 144.42095395765313, 144.67005099128556, 144.70843093515643, 147.02575214945074, 147.07009017600456, 147.48374761404892, 149.84689443310864, 150.84969099235698, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.754096
 - $\hat{q}^\beta$: 4.754096
 - q^β : 4.754096

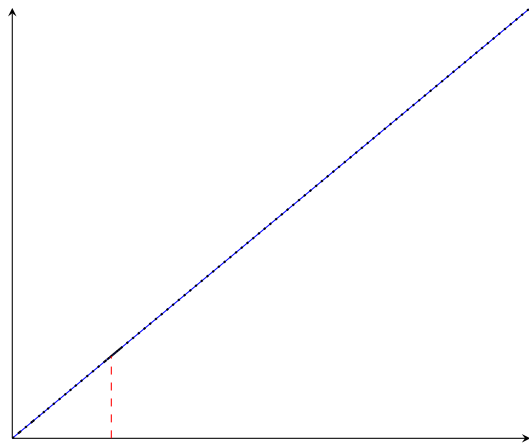
- Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.0691182452509045, 4.269280092716329, 4.432280503220532, 4.440201431981878, 4.600843803756838, 4.613047202566896, **4.707108999269609**, **4.813467676447643**, 4.825601156120121, 4.849719301830286, 4.888121952381175, 4.9192700692457025, 4.962276317638335, 4.974201802143407, 4.983938626310074, 4.993068227518925, 5.0359974771272995, 5.0418124100090695, 5.081618324442152, 5.0892956597199035, 5.195781532076625, 5.3386847474643435, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.872879
 - $\hat{q}^\beta$: 37.872879
 - q^β : 37.872879
 - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.40128299925594, 37.443498367229196, 37.47875108132638, 37.52720514437563, 37.554741031910254, 37.612770977560544, 37.63325504358228, 37.65435035996605, 37.6551299822246, 37.682669717644174, 37.71812813120871, 37.7631578546152, 37.76995088018704, 37.78573239590169, **37.821651654621526**, **37.88148462263016**, 37.9018730666985, 37.9454585315849, 37.95085342166671, 38.035748486883605, 38.10872593446926, 38.21652191748154, 38.224970554765775, 38.88483572219945, 285.4001551947793]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.321736
 - $\hat{q}^\beta$: 13.321736
 - q^β : 13.321736
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.064750414488246, 13.104034300095861, 13.134779411775817, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.195607617969486, 13.211215176722128, 13.213749911872785, 13.218222920629291, 13.22717920580809, 13.238585544475612, 13.244985069124171, 13.265751604070672, 13.266498143734903, 13.269097009439808, 13.294574825457861, **13.30871858348944**, **13.32330648750218**, 13.32663103908035, 13.346967780023082, 13.363827419897227, 13.411355634100326, 13.439500729482463, 13.468285806082731, 13.498843422507207, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



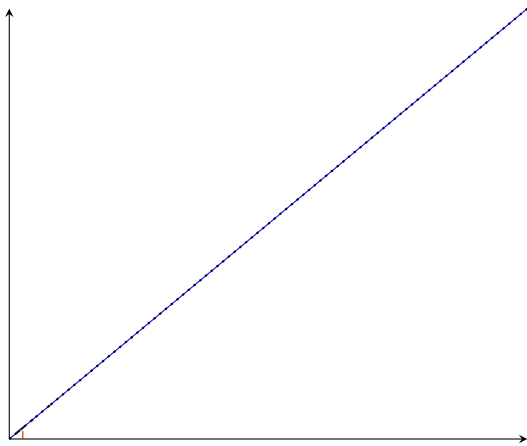
(a) $q^\beta - (1, 1, 1)$



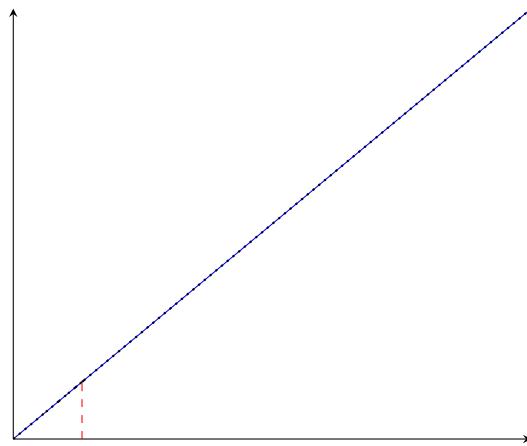
(b) $q^\beta - (1, 2, 2)$



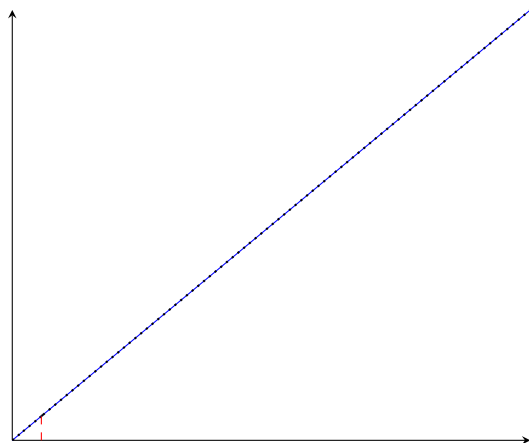
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



(a) $q^\beta - hu, 1$

7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	13.06	$b_{2,2,2}^{H,out}$	1512.4145788544556
$z_{1,2,1}$	0.0	$f_{2,2,2}^C$	0.06	$b_{1,1,2}^{C,in}$	316.3475018430171
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.013817625349338655	$b_{1,2,1}^{C,in}$	0.0
$z_{2,2,1}$	0.0	$RecLMTD_{1,2,1}$	0.01607047381606835	$b_{2,1,1}^{C,in}$	1699.5836869753139
$z_{2,2,2}$	0.0	$RecLMTD_{2,1,1}$	0.16	$b_{2,2,1}^{C,in}$	6500.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.02011172883803818	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.16	$b_{1,1,2}^{C,out}$	316.3475018430163
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	0.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.06	$b_{2,1,1}^{C,out}$	1699.5836869753139
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.06	$b_{2,2,1}^{C,out}$	6500.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.06	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	566.3202868941879		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	588.1445450882769		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	24.166666666666668	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	564.1579740786403		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	500.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	350.0		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	447.4046840943342		
$f_{1,1,2}^H$	0.7698657702980629	$b_{1,2,1}^{H,in}$	4904.144024659394		
$f_{1,2,1}^H$	7.544836961014451	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	1512.4145788544556		
$f_{2,2,2}^H$	2.563414540431282	$b_{1,1,2}^{H,out}$	447.4046840943342		
$f_{1,1,2}^C$	0.7715792727878465	$b_{1,2,1}^{H,out}$	4904.144024659382		
$f_{1,2,1}^C$	0.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	2.974602657998906	$b_{2,2,1}^{H,out}$	0.0		